S M Abu Naser Shovon

Immigration Status: Permanent Resident of the USA (Green Card Holder)
Contact: shovon.bdk@gmail.com | +1-207-992-8315 | LinkedIn | Google Scholar
Address: 625 Centennial Pkwy, Raleigh, North Carolina, USA

Professional Summary

Mechanical Engineer (BS, MS & PhD Candidate) with 5+ years of R&D work experience on UV-curable photopolymers, manufacturing process improvements, and experimental & analytical methods. Experienced in photopolymer ink/resin formulation development, materials processing & characterization, mechanical testing, 3D CAD modeling (SolidWorks), and 3D Printing (SLA, DLP, FDM, and LPBF). Skilled in collaborating with cross-functional teams, structured problem-solving methodologies (DMAIC, FMEA, RCA), and statistical data analysis. Have strong communication and interpersonal skills, and leadership ability in technical project management.

Education

North Carolina State University Aug 2022 – Present
Ph.D. in Mechanical Engineering – CGPA: 3.78/4.00
Thesis: Developing hydrogels with increased durability using digital light processing (DLP) printing for multifunctional applications
Coursework: Metal Additive Manufacturing, Finite Element Methods I & II, Polymer Science & Technology, Mechanics of Solid Polymers, Advanced Solid Mechanics

The University of Maine Aug 2019 – Aug 2022
Master of Science in Mechanical Engineering – CGPA: 3.58/4.00
Thesis: Micro-particle entrainment from a density-mismatching mixture by the dip coating process. **Link**
Coursework: Engineering Optimization, Applied Elasticity, Advanced Polymer Chemistry, Mechanics of Laminated Composites, Fluid-Structure Interaction

Bangladesh University of Engineering and Technology Feb 2013 – Sep 2017
Bachelor of Science in Mechanical Engineering – CGPA: 3.12/4.00
Coursework: Solid Mechanics, Fluid Mechanics, Thermodynamics, Heat Transfer, Heating, Ventilation & Air-Conditioning (HVAC), Applied Engineering Mathematics, Mechanics of Composite Materials

Technical Skills

Computing, Simulation, and Finite Element Analysis (FEA): MATLAB, ABAQUS
Design & Drawing: AutoCAD, SolidWorks, Mechanical Drawings, and Tolerancing (GD&T)
Mechanical Measurements, Instrumentation, and Sensing: Fluid Flow Testing, Strain Gauge Sensing (pressure, temperature, deflection), Thermal Analysis via Thermocouple, Thermistor & IR Camera
Manufacturing and Machining: Milling, Drilling, Welding, Molding, Soldering, 3D printing (SLA, DLP, FDM and LPBF), Metallurgical Sample Analysis (Cutting, Grinding, Polishing, Etching), Microstructure Analysis, Electrical Testing, Instron Mechanical Testing (Tensile, Compression, 3-Point Bending), Intense Pulsed Light (IPL) Sintering, Transient Liquid Phase (TLP) Heat Treatment
Material Characterization: SEM, UV Curing, FTIR, Digital Optical Microscopy, 3D Optical Profilometry/Confocal Microscopy (*Keyence*), Rheology, Chemical Etching, Thermography Analysis
Documentation and Statistical Data Analysis: Microsoft Office (Word, Excel, PowerPoint), Origin50, Minitab, Bio-render, ImageJ Statistical Methods (ANOVA, Regression, T-Tests, DOEs)

Core Competencies

- *Research & Development (R&D)*
- *Manufacturing Process Improvement & Optimization*
- *Material Formulation/ Recipe Developments*
- *Additive Manufacturing (AM) or 3D Printing*
- *Statistical Analysis & Experimental Designs (DOEs)*
- *Materials Characterization & Mechanical Testing*
- *Technical Project Leadership & Management*
- *Failure Root-Cause Analysis (FMEA & RCA)*

Professional Experiences

Graduate Assistant at North Carolina State University, Raleigh, NC, USA Aug 2022 – Present

- Developed tough hydrogel formulations for DLP-based additive manufacturing, conducting mechanical testing, process optimization, fabrication, and microscopy.
- Engineered hydrogels with additive fillers for optical and wearable sensor applications using DLP 3D printing, mechanical testing, FTIR, SEM, and microscopy.

- Formulated visible-light and UV-curable polymer inks including hydrogels, elastomers, ionic, and photopolymer systems.
- Fabricated flexible electronics sensors via hybrid micro-nano manufacturing combining IPL sintering, acoustic patterning, and UV curing, characterized via microscopic analysis.
- Developed a dual-network & dual-crosslinked polymer-based ionically conductive hydrogel via Continuous DLP (CDLP) 3D printing and performed performance analysis using Instron mechanical testing, microscopy, and electrical resistance analysis.
- Developed and characterized a framework of acoustically aligned and bulk nanocomposites' performance in UV curing & IPL sintering and analyzed the performance in electrical resistance analysis.

Visiting Researcher at Technical University of Dresden, Germany May 2025 – July 2025

- Applied ultrasound actuation in meta-structures and investigated heat-transfer variation with the change of 3D-printed design.
- Implemented the efficient meta-structure design with LCEs and performed the ultrasound actuation of LCEs.

Graduate Assistant at University of Maine, Orono, ME, USA *Sep 2019 – Jul 2022*

- Developed thin-film processing through data-driven experimentation, root-cause analysis, and statistical analysis (DOE, ANOVA) using SEM, FTIR, 3D imaging, and rheological characterization.
- Optimized the thin-film deposition & particle suspension formulation by tuning the MW of feed polymer and viscosity and characterized the formulation by SEM, rheology, and 3D imaging.
- Investigated capillary joining and interlayer transient liquid phase (TLP) bonding of metal rod structures through metallographic sample preparation, microstructural analysis, and digital 3D microscopy.

Graduate Assistant at North Carolina State University, Raleigh, NC, USA *Aug 2022 – Dec 2026*

- Developed course materials for taking classes, led the senior design project sponsored by USDA and Corning among teams, and assessed the student performance in projects and reports.
- Mentored student teams through full project lifecycle: problem definition, design, prototyping, testing, and presentation.
- Assessed student performance through structured design reviews, reports, and presentations; managed project logistics including budget procurement and lab resource coordination.

Presentations and Posters

- *3D Printing of Hydrogels via Continuous Digital Light Processing (CDLP) and Simulation for Dimensional Accuracy and Structural Stability* – Presented at ASME Manufacturing Science and Engineering Conference (MSEC 2026)
- *3D Fabrication of NIR Fluorescent SWCNT Hydrogels via Continuous Digital Light Processing* - Presented at NC State Institute of Connected Sensor-Systems (IConS) Fall Meeting 2025
- *Grayscale Variation and Multifunctional Performance Analysis of Hydrogels utilizing Continuous Digital Light Processing System* - Presented at NC State MAE Symposium 2025s
- *Effect of Surfactants on the Acoustic Patterning of organic and inorganic Microparticles in a Polymer Matrix* - Presented at IMECE 2024
- *Development of hydrogel ink for DLP printing system and characterizing the hydrogels for application on hydrated environment* - Presented at NC State MAE Symposium 2024
- *Inorganic micro-particle transfer from liquid carrier system by dipping* - Presented at Society of Rheology 92nd Meeting 2021.
- *Solid transfer of large particles by dipping in a heterogeneous mixture* - Presented at SES 2020, MSEC 2021, University of Maine Student Symposium 2021.
- *Effect of Withdrawal Velocity on Particle Entrainment from Density Mismatched Mixture* - Presented at MSEC 2022

Projects

- **Investigation of Powder Characteristics and Microstructural Evolution of Laser Printed Ti-15Mo Samples Prepared by Ball Milling:** Prepared Titanium-Molybdenum (Ti-15Mo) alloy using Ball Milling, characterized the powder via SEM, LECO, Optical Microscopy, and printed the powder using Laser Powder Bed Fusion (LPBF).
- **Analysis of Cylindrical Bending of a Symmetric Sandwich Composite Plate:** The maximum deflection and failure analysis of the plate were performed through analytical derivations and compared with the finite element analysis using ABAQUS.

Professional Referee

- **Dr. Henry Ware**, Assistant Professor, North Carolina State University, Email: howare@ncsu.edu
- **Dr. Chau M. Tran**, Associate Professor, North Carolina State University, Email: cmtran@ncsu.edu